

Theorem: $MIP(\delta = 1/3s) \Rightarrow NSP(s)$ ^{سببنا} $|S| = s$

Proof:

$$\theta_s^* \in \text{null}(A): \left| \theta_s^{*T} \left(\frac{A^T A}{n} - I \right) \theta_s^* \right| \leq \theta_s^* \begin{matrix} s \\ \left\{ \begin{bmatrix} \theta^* \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\} \\ s^c \end{matrix}$$

فرض $v^T X v = \sum v_i v_j x_{ij}$ $\underbrace{\|A^T A/n - I\|_\infty}_\delta \|\theta_s^*\|_1^2$

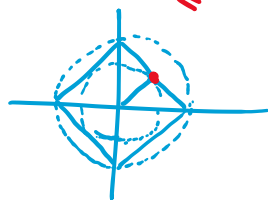
$$|v^T X v| = \left| \sum v_i v_j x_{ij} \right| \leq \sum |v_i| |v_j| |x_{ij}|$$

$$\leq \|X\|_\infty \sum |v_i| |v_j|$$

$$= \|X\|_\infty \|v\|_1^2$$

فرض $\theta^* \in \text{null}(A)$ $A \begin{bmatrix} \theta_s^* \\ 0 \end{bmatrix} = A \begin{bmatrix} 0 \\ -\theta_{s^c}^* \end{bmatrix}$

$$\underbrace{\theta_s^{*T} \left(-\frac{X^T X}{n} + I \right) \theta_s^*}_{\|\theta_s^*\|_2^2 - \theta_s^{*T} \frac{X^T X}{n} \theta_s^*} \leq \left| \theta_s^{*T} \left(\frac{X^T X}{n} - I \right) \theta_s^* \right| \leq \delta \underbrace{\|\theta_s^*\|_1^2}_{\leq \delta s \|\theta_s^*\|_2^2}$$



فرض $\frac{1}{\sqrt{d}} \|v\|_1 \leq \|v\|_2 \leq \|v\|_1$

$$\|v\|_1 = \langle v, \text{sgn}(v) \rangle \leq \|v\|_2 \cdot \underbrace{\|\text{sgn}(v)\|_2}_{\sqrt{d}}$$

$$\|v\|_2 \geq \frac{1}{\sqrt{d}} \|v\|_1$$

$$\underbrace{(1-\delta\delta)}_s \|\theta_s^*\|^2 \leq \theta_s^{*T} \left(\frac{A^T A}{n} \right) \theta_s^*$$

$$\geq \frac{1-\delta\delta}{s} \|\theta_s^*\|^2$$

$$\frac{1-\delta\delta}{s} \|\theta_s^*\|_1 \leq \theta_s^{*T} \left(\frac{A^T A}{n} \right) \theta_s^*$$

$$= -\theta_s^{*T} \frac{A^T A}{n} \theta_{s^c}^*$$

$$\leq \left| \theta_s^{*T} \frac{A^T A}{n} \theta_{s^c}^* \right|$$

$$= \left| \theta_s^{*T} \left(\frac{A^T A}{n} - I \right) \theta_{s^c}^* \right|$$

$$\leq \underbrace{\left\| \frac{A^T A}{n} - I \right\|_\infty}_\delta \cdot \cancel{\|\theta_s^*\|_1} \cdot \|\theta_{s^c}^*\|_1$$

$$\Rightarrow \|\theta_s^*\|_1 \underbrace{\left(\frac{1-\delta\delta}{\delta\delta} \right)}_s \leq \|\theta_{s^c}^*\|_1$$

$$\|\theta_s^*\|_1 \leq \left(\frac{\delta\delta}{1-\delta\delta} \right) \|\theta_{s^c}^*\|_1$$

It's sufficient to take $\frac{\delta\delta}{1-\delta\delta} \leq \frac{1}{2}$

$$2\delta\delta \leq 1-\delta\delta \Rightarrow 3\delta\delta \leq 1 \Rightarrow \boxed{\delta \leq \frac{1}{3s}}$$

$$\frac{\delta\delta}{1-\delta\delta} \leq 1-\epsilon$$

□

1h. - i.i.d Rad. Then $n \geq O(s^2 \log d)$

A satisfies $S^2(S)$ $|S|=s$ with high prob.

$$\mathbb{P} \left(\max_{i,j} \left| \frac{\langle A_i, A_j \rangle}{n} - \delta_{ij} \right| \geq \frac{1}{3}s \right) \leq \epsilon$$

$$\mathbb{P} \left(\max C_i \geq t \right) \leq \mathbb{P} \left(C_1 \geq t \cup \dots \cup C_N \geq t \right) \leq \sum \mathbb{P}(C_i \geq t)$$

$$\max C_i \geq t \Rightarrow \exists i \quad C_i \geq t$$

$$\underbrace{C_1 \geq t \cup C_2 \geq t \dots \cup C_N \geq t}$$

$$C_{ij} = \left| \frac{\langle A_i, A_j \rangle}{n} - \delta_{ij} \right| \quad i \neq j$$

$$\underbrace{\frac{1}{n} \sum_{k=1}^n A_{ik} A_{jk}}_{X_k \text{ i.i.d Rad}} \rightarrow \mathbb{E}(X_k)$$

$$\mathbb{P}(C_{ij} \geq t) \leq 2 e^{-\frac{nt^2}{2}}$$

$$\Rightarrow \mathbb{P} \left(\max_{i,j} C_{ij} \geq \frac{t}{\frac{1}{3}s} \right) \leq \underbrace{2 \binom{d}{2} e^{-\frac{nt^2}{2}}}_{\leq \epsilon} e^{-\frac{n}{18s^2}} \leq d^2 \dots$$

$$d^2 e^{-\frac{n}{18s^2}} \leq \epsilon$$

$$\log d^2 - \frac{n}{18s^2} \leq \log \epsilon \Rightarrow n \geq \underbrace{18 s^2 \log d^2 / \epsilon}_{O(s^2 \log d)}$$